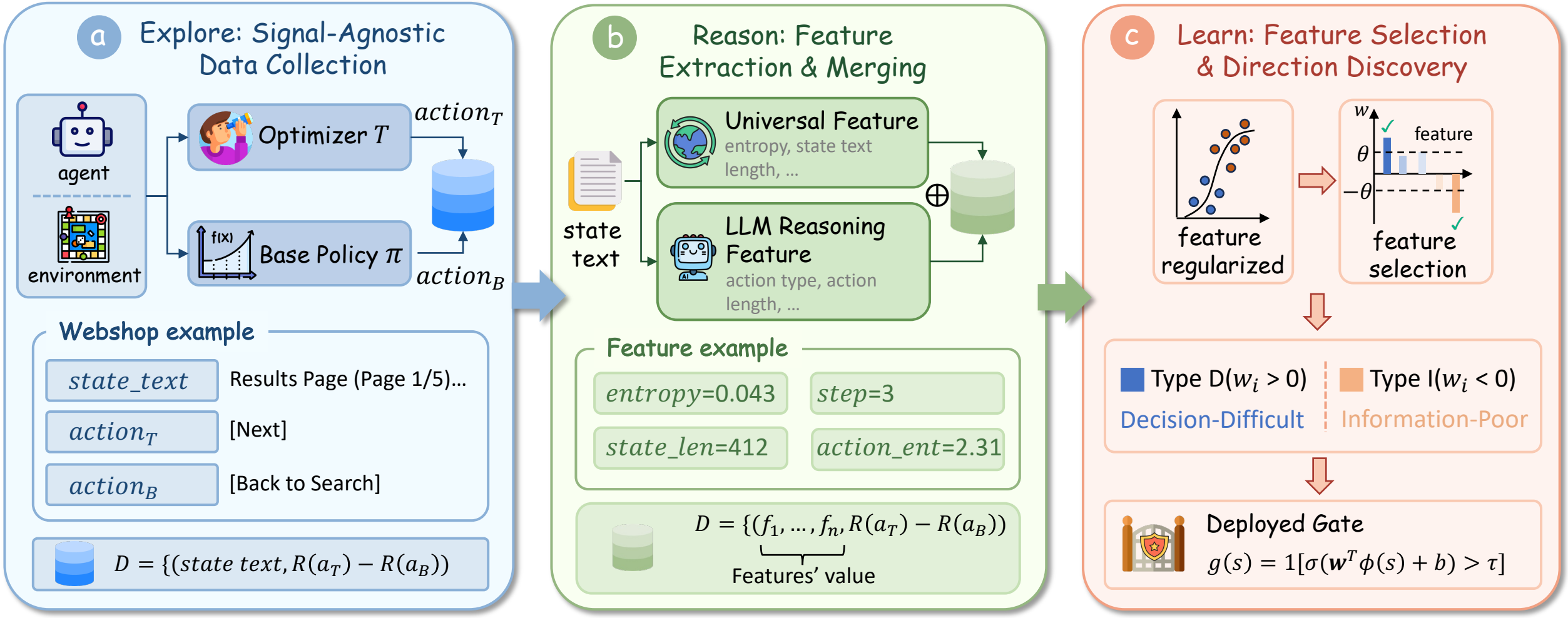
a
Explore: Signal-Agnostic Data Collection
agent
environment
Optimizer T
Base Policy π
$action_T$
$action_B$
Webshop example
$state_text$
Results Page (Page 1/5)...
$action_T$
[Next]
$action_B$
[Back to Search]
$D = \{(state\ text, R(a_T) - R(a_B))$
b
Reason: Feature Extraction & Merging
state text
Universal Feature
entropy, state text length, ...
LLM Reasoning Feature
action type, action length, ...
Feature example
$entropy$=0.043
$step$=3
$state_len$=412
$action_ent$=2.31
$D = \{(f_1, \dots, f_n, R(a_T) - R(a_B))$
Features' value
c
Learn: Feature Selection & Direction Discovery
feature regularized
feature selection
w
feature
θ
$-\theta$
Type D($w_i > 0$)
Decision-Difficult
Type I($w_i < 0$)
Information-Poor
Deployed Gate
$g(s) = 1[\sigma(\mathbf{w}^T\phi(s) + b) > \tau]$